

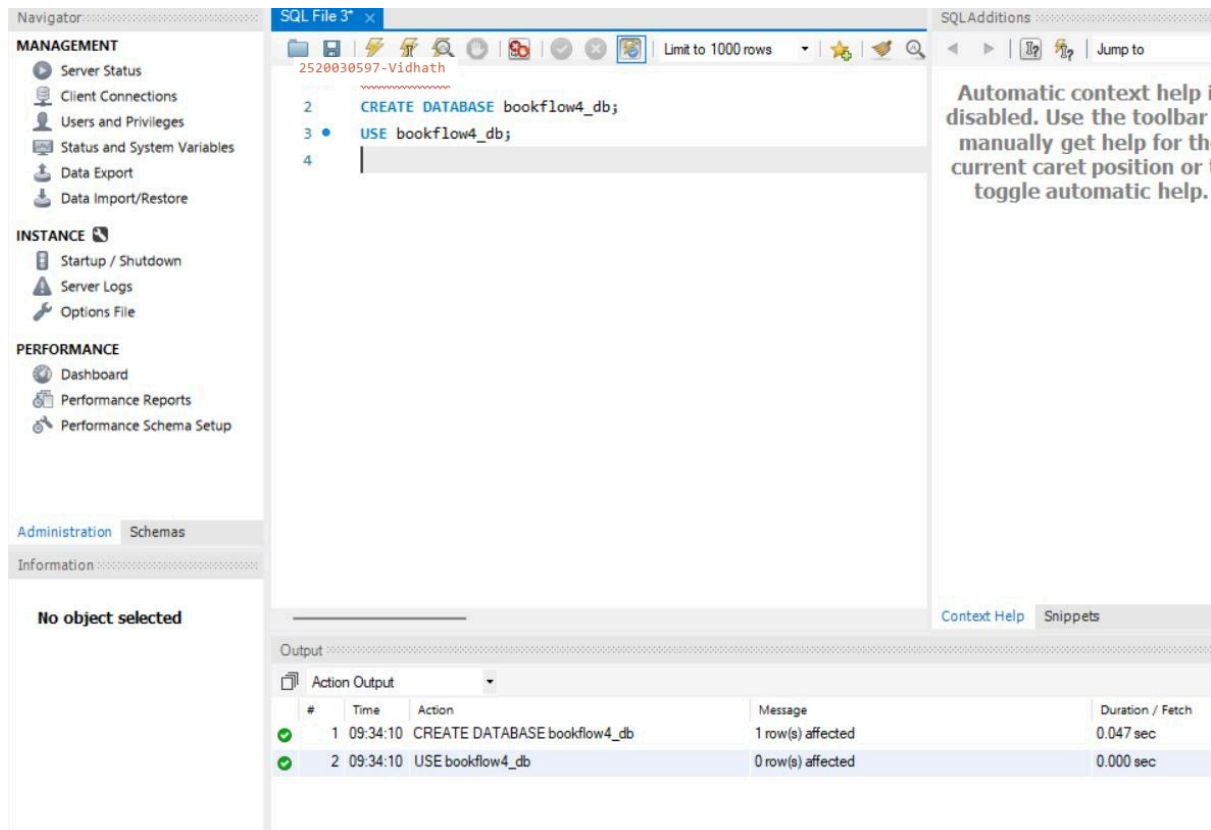
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030597

NAME: U.VIDHATH

Query 1: -CREATE DATABASE



The screenshot displays the SQL Enterprise Manager interface. The left pane shows the 'Navigation' tree with categories like MANAGEMENT, INSTANCE, and PERFORMANCE. The central pane shows the SQL script being executed:

```
2 CREATE DATABASE bookflow4_db;
3 USE bookflow4_db;
4
```

The right pane shows the 'SQLAdditions' tab with a message: 'Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.'

The bottom pane shows the 'Output' window with the 'Action Output' tab selected. It displays the results of the executed queries:

#	Time	Action	Message	Duration / Fetch
1	09:34:10	CREATE DATABASE bookflow4_db	1 row(s) affected	0.047 sec
2	09:34:10	USE bookflow4_db	0 row(s) affected	0.000 sec

Query 2: Create Books Table (DDL + Constraints)



The screenshot displays the SQL Enterprise Manager interface. The left pane shows the 'Navigation' tree. The central pane shows the SQL script being executed:

```
5 CREATE TABLE BOK (
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK (published_year < 2027)
10 );
```

The right pane shows the 'SQLAdditions' tab with a message: 'manually get help for the current caret position or toggle automatic help.'

Output: -

✓	4	09:38:44	CREATE TABLE Bok (book_id INT PRIMARY KEY...	0 row(s) affected	0.109 sec
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Query 3: Insert Books and display

```
11 2520030597-Vidhath
12 INSERT INTO Bok (book_id,title,isbn,published_year) VALUES
13 (1,'The Great Bay','9780743273565',2000),
14 (2,'The Indian Culture',2345987456,2010),
15 (3,'1947','9456783457934',2020);
16 SELECT * FROM Bok;
```

Automatic context menu disabled. Use the left mouse button to manually get help for the current caret position or toggle automatic help.

Result Grid

book_id	title	isbn	published_year
1	The Great Bay	9780743273565	2000
2	The Indian Culture	2345987456	2010
3	1947	9456783457934	2020
NULL	NULL	NULL	NULL

Form Editor

Field Types

Output:-

✓	12	12:10:49	INSERT INTO Bok (book_id,title,isbn,published_y...	3 row(s) affected	Records: 3 Duplicates: 0 Wami...	0.062 sec
✓	13	12:10:49	SELECT * FROM Bok LIMIT 0, 1000	3 row(s) returned		0.000 sec / 0.000 sec

Query 4: Members Table: Create

```
CREATE TABLE Members(
  member_id INT PRIMARY KEY,
  full_name VARCHAR(100),
  email VARCHAR(100) UNIQUE
);
```

Output:-

✓	19	12:20:46	CREATE TABLE Members(member_id INT PRIM...	0 row(s) affected		0.063 sec
---	----	----------	--	-------------------	--	-----------

Query 5:- Insert Members table

```
INSERT INTO Members (member_id,full_name,email) VALUES
(101, 'Vidhath', 'vidhath@gmail.com'),
(102, 'Raghuveer', 'raghuveer@gmail.com'),
(103, 'Durga prasad', 'durgaprasad@gmail.com');
```

Output: -

20	12:22:31	INSERT INTO Members (member_id,full_name,e...	3 row(s) affected Records: 3 Duplicates: 0 Wami...	0.016 sec
----	----------	---	--	-----------

Query 6: - : Display Members table

```
SELECT * FROM Members;
```

Output: -

The screenshot shows a database management interface. At the top, there's a 'Result Grid' tab. Below it, a table with 3 columns: 'member_id', 'full_name', and 'email'. The table contains 3 rows of data:

member_id	full_name	email
101	Vidhath	vidhath@gmail.com
102	Raghuveer	raghuveer@gmail.com
103	Durga prasad	durgaprasad@gmail.com

Below the table, there's a 'Members 2' tab. At the bottom, there's an 'Output' section showing a log of queries and their results:

Query ID	Timestamp	Query	Result	Duration
20	12:22:31	INSERT INTO Members (member_id,full_name,e...	3 row(s) affected Records: 3 Duplicates: 0 Wami...	0.016 sec
21	12:24:30	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query 7: Create Loans Table

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```
> CREATE TABLE Loan (  
  loan_id INT PRIMARY KEY,  
  member_id INT,  
  book_id INT,  
  loan_date DATE,  
  FOREIGN KEY (member_id) REFERENCES Members(member_id),  
  FOREIGN KEY (book_id) REFERENCES Bok(book_id)  
);
```

Output:-

```
23 12:30:52 CREATE TABLE Loan (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN... 0 row(s) affected
```

Query 8: Insert 10 Loans

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Output: -

```
91 13:40:13 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'), (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'), (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'), (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01'); 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
```

Query 9: Display

```
SELECT * FROM Loans;
```

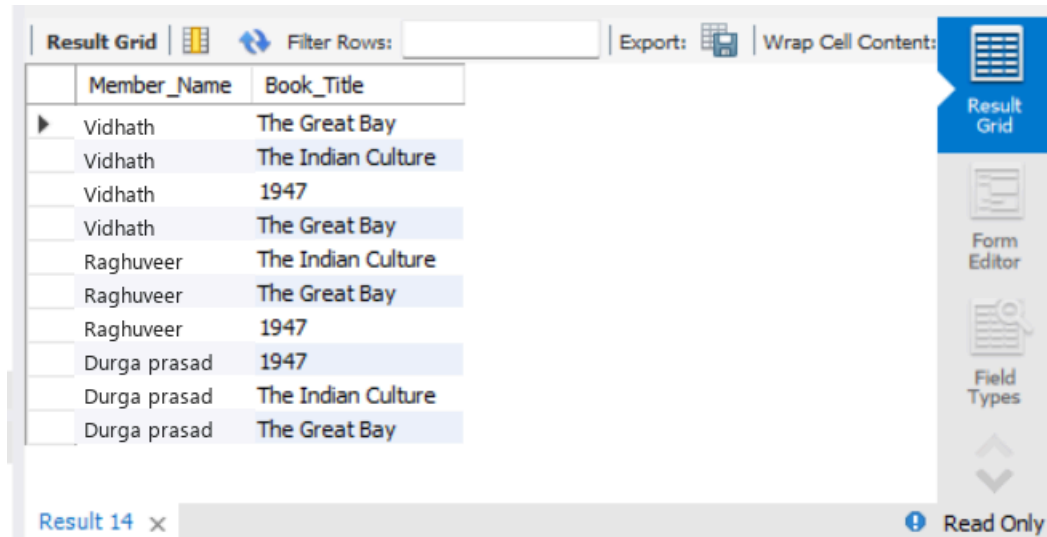
Output :-

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
*	NULL	NULL	NULL	NULL

Query 10: JOIN Query

```
45 • SELECT m.full_name AS Member_Name,b.title AS Book_Title
46 FROM loans l
47 INNER JOIN Members m ON l.member_id = m.member_id
48 INNER JOIN Bok b ON l.book_id = b.book_id;
```

Output: -



Member_Name	Book_Title
Vidhath	The Great Bay
Vidhath	The Indian Culture
Vidhath	1947
Vidhath	The Great Bay
Raghuveer	The Indian Culture
Raghuveer	The Great Bay
Raghuveer	1947
Durga prasad	1947
Durga prasad	The Indian Culture
Durga prasad	The Great Bay

✓ 103 13:50:44 SELECT m.full_name AS Member_Name,b.title A... 10 row(s) returned

Query 11: GROUP BY Query

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Bok
GROUP BY published_year
ORDER BY published_year;
```

Output :-

Result Grid			Filter Rows:	Export:	Wrap Cell Contents:
	published_year	Total_Books			
▶	2000	1			
	2010	1			
	2020	1			

✓	103	13:50:44	SELECT m.full_name AS member_name, b.title AS...	10 row(s) returned	0.000 sec / 0.000 sec
✓	104	13:54:20	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned	0.016 sec / 0.000 sec

Query 12: Donation via TRANSACTION

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```
CREATE TABLE Donation_History (
donation_id INT PRIMARY KEY,
book_id INT,
donar_name VARCHAR(100),
donation_date DATE,
FOREIGN KEY (book_id) REFERENCES Bok(book_id)
);
```

Output: -

✓	107	13:57:39	CREATE TABLE Donation_History (donation_id I...	0 row(s) affected	0.094 sec
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Query 13:

```
START TRANSACTION;
INSERT INTO BOK (book_id,title,isbn,published_year)
VALUES (4,'Animal Farm','9978654321980',1945);
INSERT INTO Donation_History(donation_id,book_id,donar_name,donation_date)
VALUES(1,4,'Raj Kumar',CURDATE());
COMMIT;
```

Output: -

✓	109	14:02:19	COMMIT	0 row(s) affected
✓	110	14:02:25	START TRANSACTION	0 row(s) affected
✓	111	14:02:29	INSERT INTO BOK (book_id,title,isbn,published_...	1 row(s) affected

✓ 113 14:02:50 INSERT INTO Donation_History(donation_id,boo... 1 row(s) affected

Query 14: Index on ISBN

```
CREATE INDEX idx_books_isbn ON Bok(isbn);
```

Output :-

✓ 116 14:05:25 CREATE INDEX idx_books_isbn ON Bok(isbn) 0 row(s) affected Records: 0 Duplicates: 0 Wami... 0.125 sec

Query 15: Fast Search

```
SELECT * FROM BOK WHERE isbn = '9780743273565';
```

Output : -

The screenshot shows a database application interface. At the top, there's a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. Below the toolbar is a table with the following data:

	book_id	title	isbn	published_year
▶	1	The Great Bay	9780743273565	2000
*	NULL	NULL	NULL	NULL

On the right side of the table, there's a vertical sidebar with buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom of the application window, there's a status bar that says 'BOK 16' and 'Apply'. Below the application window, there's a query log showing the execution of the search query:

✓ 117 14:07:50 SELECT * FROM BOK WHERE isbn = '97807432... 1 row(s) returned